

E.11 Chapter 12

Django was originally built for the Web 1.0 world. To add more interactivity, you can use a heavy JavaScript framework, like React, or a lighter-weight library, like HTMX. HTMX provides API-like functionality through HTML attributes, dynamically replacing portions of a page.

Exercise 1

Add search functionality for your ad content similar to the musicians search page. Implement both search-as-you-type and infinite scrolling.

The HTMX `hx-get` attribute interacts with an input box by appending its contents to the URL's query parameter. This can be used to build a search-as-you-type feature. The following listing contains the HTML for an ad-searching page.

Listing E.51 HTML for a search ads page

```
<!-- RiffMates/templates/search_ads.html -->
{% extends "base.html" %}

{% block content %}
  <h1>Search Ads</h1>

  <input name="search_text"
    placeholder="Search the ads..."
    type="text"
    value="{{ search_text }}"
    hx-get="{% url 'search_ads' %}"
    hx-trigger="keyup changed delay:500ms"
    hx-target="#results"
    hx-push-url="true">

  <div class="results" id="results">
    {% include "partials/ad_results.html" %}
  </div>
{% endblock content %}
```

When new content is fetched, put it in the <div> with ID results.

To support deep linking, populate the input box with any search term given to the page at render time.

The hx-get attribute specifies the URL to call when content is added to the input box.

The hx-trigger attribute dictates when the GET is performed. This value is based on a keyup event but only if it changes the contents of the input box and occurs 500 ms after the last press.

Append the contents of the input box to the query parameter in the browser's nav to enable deep linking and bookmarking.

The <div> tag that is the target of results

To support deep linking, include the subtemplate that populates any results available at render time.

Following the common pattern in chapter 12, the HTML is broken into two parts: the main page and the partial template used to render the results. The partial template displays the ad search results.

To implement infinite scroll on the search results, the search data gets paginated. The partial that displays the results also needs to contain the code that triggers fetching more data. It does this by using a value of `revealed` for the `hx-trigger` attribute, calling for the next page of data to display. The following listing shows the complete HTML for displaying a partial, with both the page of search results as well as the code for triggering the fetch action for next chunk.